School of Future Tech

Case Study Report

ON

<u>BioGazer – DNA Analysis and Mutation Tracking System</u>

BY

PRINCE SINGH

ROLL NO: 150096725168

Problem Statement :-

Modern genetic research involves analyzing large DNA sequences and identifying mutations efficiently. Researchers require a platform that can visualize DNA sequences, search genetic information, manage patient records, and assess mutation risks through an interactive interface.

BioGazer is a React.js-based DNA Analysis and Visualization System that enables scientists to view, edit, validate, and analyze genetic sequences. The platform provides color-coded DNA visualization, mutation tracking, gene searching, risk analysis, patient record management, and customizable settings through a modern dashboard interface.

Tech Stack :-

Frontend Technologies

- React.js
- JavaScript
- HTML
- CSS

Libraries and Tools

- React Router DOM
- Axios
- Vite

React Concepts Used

- Functional Components
- useState
- useEffect
- useContext
- useMemo

- useCallback
- Props
- Conditional Rendering
- Lists and Keys
- Context API

Development Tools

- Visual Studio Code
- GitHub
- Vercel

Component Architecture :-

The project follows a component-based architecture to improve scalability, maintainability, and reusability.

Main Components

- **Navbar**
- **Sidebar**
- **Layout**
- **StatCard**
- **DNA Strand Viewer**
- **Patient Table**
- **Pagination**
- **Risk Badge**
- **Loader**
- **Empty State**

Pages

- **Home Dashboard**
- **DNA Viewer**
- **Patients**
- **Gene Search**
- **Risk Calculator**

- Settings

Context Management

AppContext is used to manage:

- Theme Settings
- Notification Settings
- DNA Display Settings
- Global Application State

Routing Structure

```
App Component
 │
 ├── Home
 ├── DNA Viewer
 ├── Patients
 ├── Gene Search
 ├── Risk Calculator
 └── Settings
```

Features :-

1. Dashboard

Displays project statistics, DNA information, and quick navigation options.

2. Color-Coded DNA Viewer

Visualizes DNA sequences using different colors for A, T, G, and C nucleotides.

3. DNA Mutation Tracking

Identifies and highlights mutations in DNA sequences.

4. Gene Editing with Undo/Redo

Allows users to edit DNA sequences and restore previous changes.

5. DNA Sequence Organizer

Provides search, sorting, and filtering of DNA records.

6. Patient Records Management

Manages and displays 1200+ patient records efficiently.

7. Mutation Risk Calculator

Calculates and displays mutation risk levels based on mutation count.

8. Smart Gene Search

Performs real-time searching and filtering of gene information.

9. DNA Validation System

Validates DNA input and accepts only valid nucleotide characters.

10. Study Settings Hub

Allows users to customize themes, notifications, and application settings.

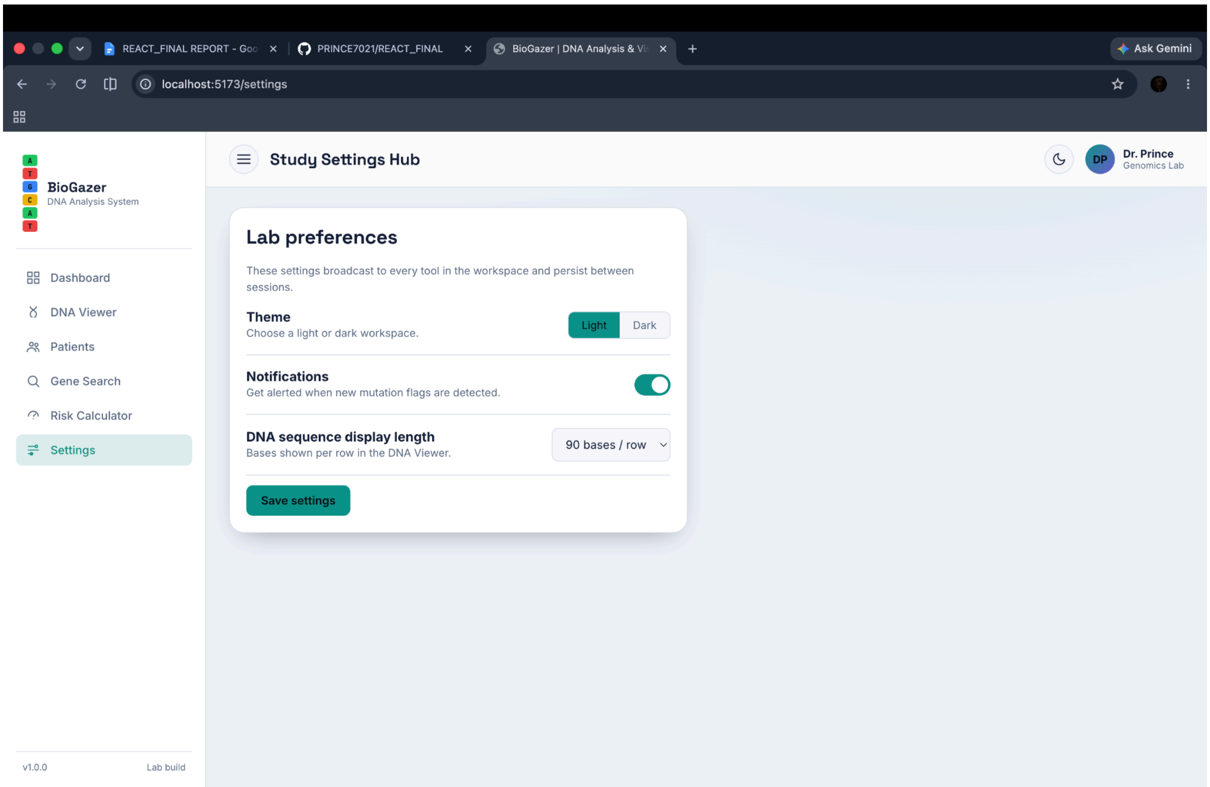
11. Performance Optimization

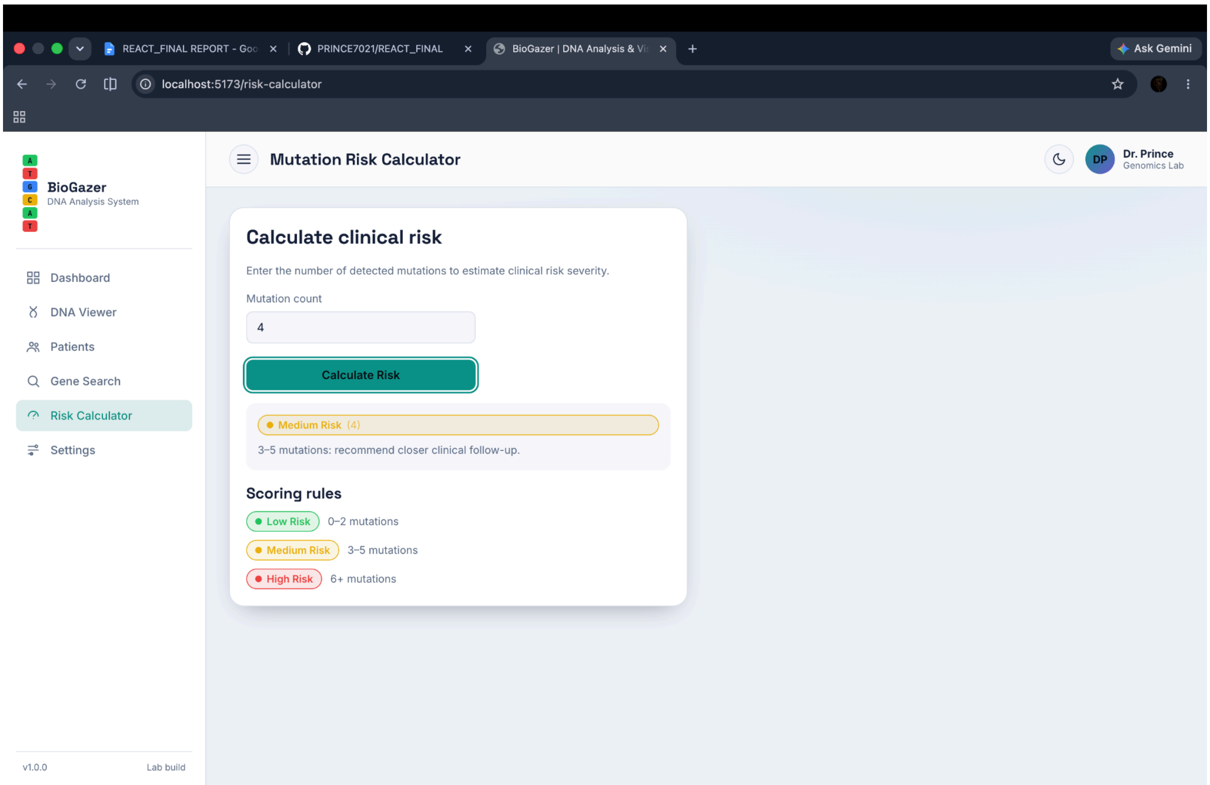
Uses pagination and React optimization techniques for smooth performance.

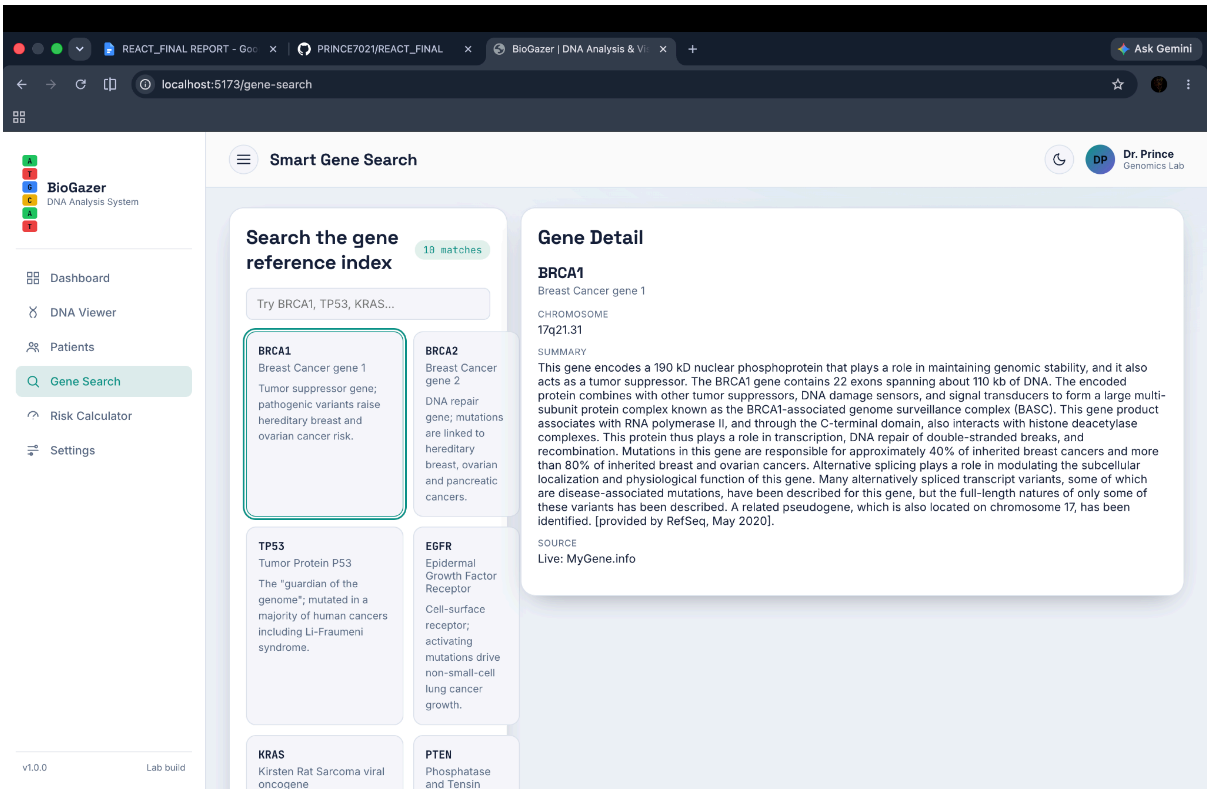
12. Responsive Design

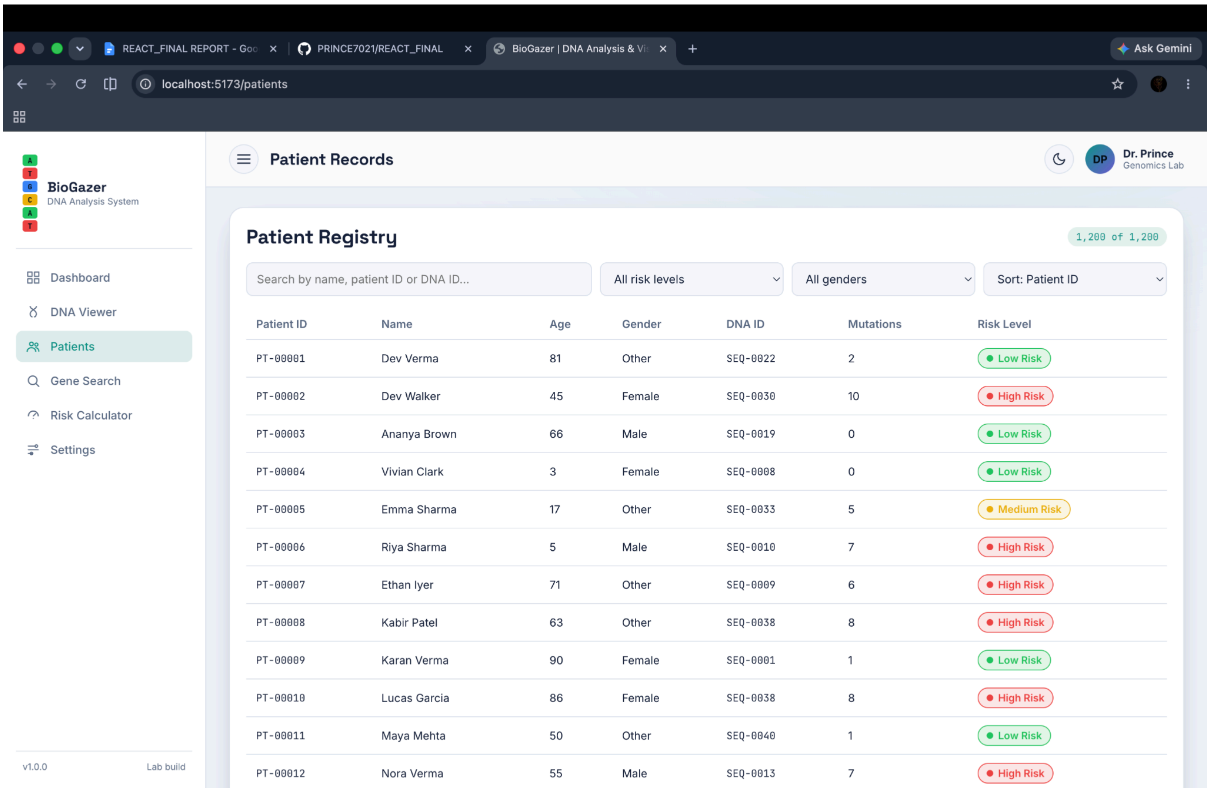
Provides a user-friendly interface across desktop, tablet, and mobile devices.

Screenshots :-






REACT_FINAL REPORT - Goo
PRINCE7021/REACT_FINAL
BioGazer | DNA Analysis & Vi
Ask Gemini
localhost:5173/gene-search
BioGazer
DNA Analysis System
Dashboard
DNA Viewer
Patients
Gene Search
Risk Calculator
Settings
v1.0.0
Lab build
Smart Gene Search
Dr. Prince
Genomics Lab
Search the gene reference index
10 matches
Try BRCA1, TP53, KRAS...
BRCA1
Breast Cancer gene 1
Tumor suppressor gene; pathogenic variants raise hereditary breast and ovarian cancer risk.
BRCA2
Breast Cancer gene 2
DNA repair gene; mutations are linked to hereditary breast, ovarian and pancreatic cancers.
TP53
Tumor Protein P53
The "guardian of the genome"; mutated in a majority of human cancers including Li-Fraumeni syndrome.
EGFR
Epidermal Growth Factor Receptor
Cell-surface receptor; activating mutations drive non-small-cell lung cancer growth.
KRAS
Kirsten Rat Sarcoma viral oncogene
PTEN
Phosphatase and Tensin
Gene Detail
BRCA1
Breast Cancer gene 1
CHROMOSOME
17q21.31
SUMMARY
This gene encodes a 190 kD nuclear phosphoprotein that plays a role in maintaining genomic stability, and it also acts as a tumor suppressor. The BRCA1 gene contains 22 exons spanning about 110 kb of DNA. The encoded protein combines with other tumor suppressors, DNA damage sensors, and signal transducers to form a large multi-subunit protein complex known as the BRCA1-associated genome surveillance complex (BASC). This gene product associates with RNA polymerase II, and through the C-terminal domain, also interacts with histone deacetylase complexes. This protein thus plays a role in transcription, DNA repair of double-stranded breaks, and recombination. Mutations in this gene are responsible for approximately 40% of inherited breast cancers and more than 80% of inherited breast and ovarian cancers. Alternative splicing plays a role in modulating the subcellular localization and physiological function of this gene. Many alternatively spliced transcript variants, some of which are disease-associated mutations, have been described for this gene, but the full-length natures of only some of these variants has been described. A related pseudogene, which is also located on chromosome 17, has been identified. [provided by RefSeq, May 2020].
SOURCE
Live: MyGene.info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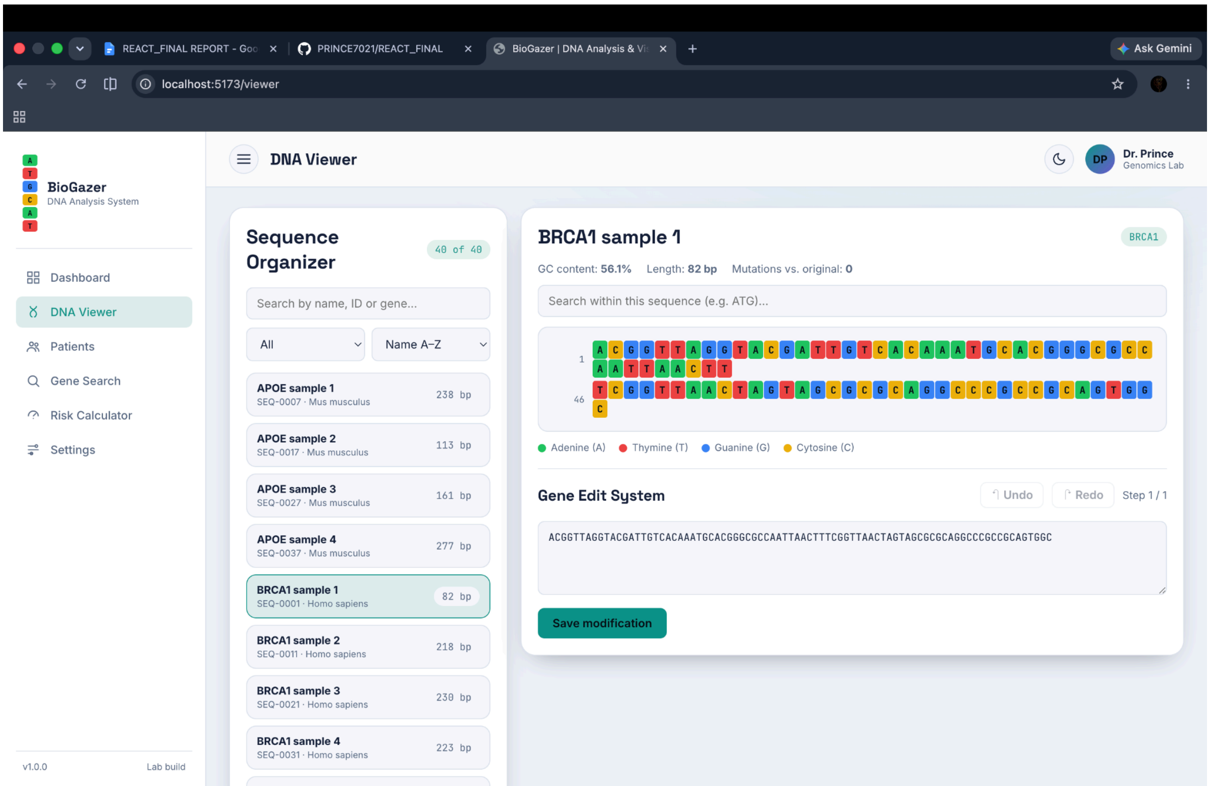
DNA Analysis System
Dashboard
DNA Viewer
Patients
Gene Search
Risk Calculator
Settings
v1.0.0
Lab build
Patient Records
Dr. Prince
Genomics Lab
Patient Registry
1,200 of 1,200
Search by name, patient ID or DNA ID...
All risk levels
All genders
Sort: Patient ID
Patient ID Name Age Gender DNA ID Mutations Risk Level
PT-00001 Dev Verma 81 Other SEQ-0022 2 Low Risk
PT-00002 Dev Walker 45 Female SEQ-0030 10 High Risk
PT-00003 Ananya Brown 66 Male SEQ-0019 0 Low Risk
PT-00004 Vivian Clark 3 Female SEQ-0008 0 Low Risk
PT-00005 Emma Sharma 17 Other SEQ-0033 5 Medium Risk
PT-00006 Riya Sharma 5 Male SEQ-0010 7 High Risk
PT-00007 Ethan Iyer 71 Other SEQ-0009 6 High Risk
PT-00008 Kabir Patel 63 Other SEQ-0038 8 High Risk
PT-00009 Karan Verma 90 Female SEQ-0001 1 Low Risk
PT-00010 Lucas Garcia 86 Female SEQ-0038 8 High Risk
PT-00011 Maya Mehta 50 Other SEQ-0040 1 Low Risk
PT-00012 Nora Verma 55 Male SEQ-0013 7 High Risk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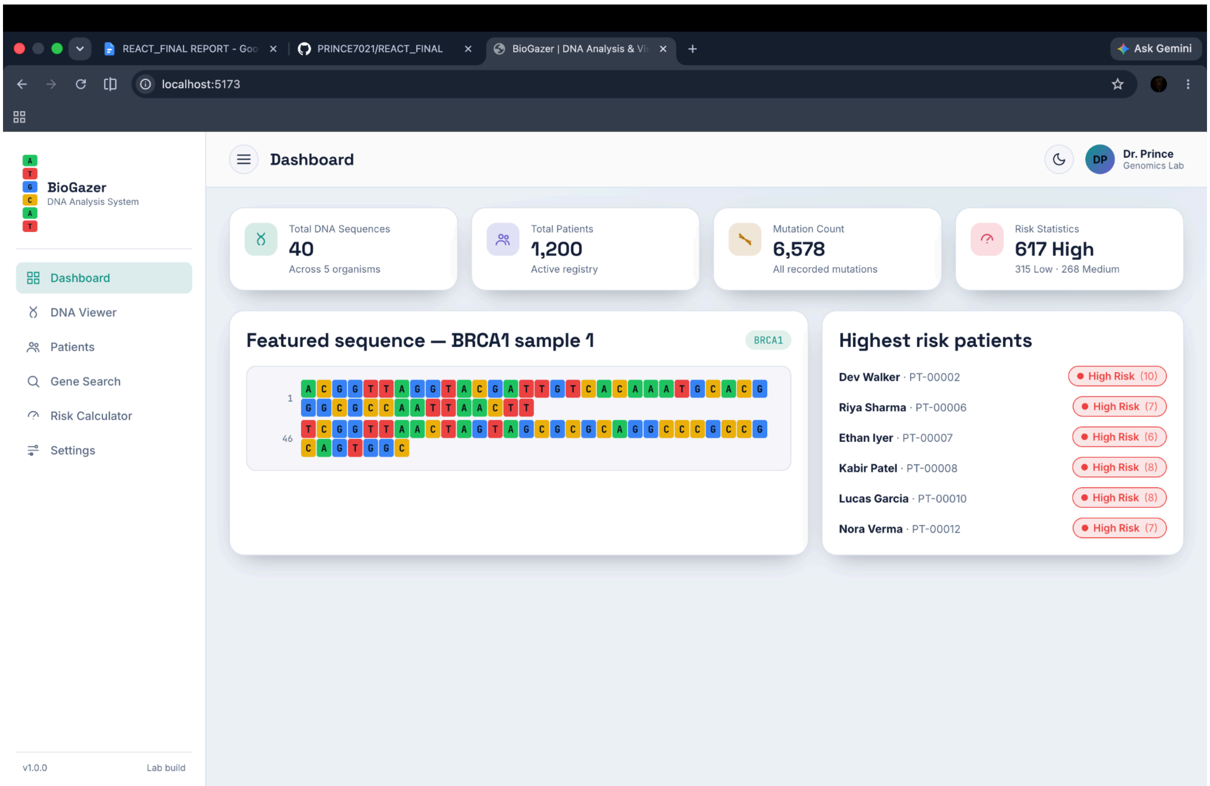
v1.0.0
Lab build
DNA Viewer
Dr. Prince
Genomics Lab
Sequence Organizer
40 of 40
Search by name, ID or gene...
All
Name A–Z
APOE sample 1
SEQ-0007 · Mus musculus
238 bp
APOE sample 2
SEQ-0017 · Mus musculus
113 bp
APOE sample 3
SEQ-0027 · Mus musculus
161 bp
APOE sample 4
SEQ-0037 · Mus musculus
277 bp
BRCA1 sample 1
SEQ-0001 · Homo sapiens
82 bp
BRCA1 sample 2
SEQ-0011 · Homo sapiens
218 bp
BRCA1 sample 3
SEQ-0021 · Homo sapiens
230 bp
BRCA1 sample 4
SEQ-0031 · Homo sapiens
223 bp
BRCA1 sample 1
BRCA1
GC content: 56.1% Length: 82 bp Mutations vs. original: 0
Search within this sequence (e.g. ATG)...
Adenine (A) Thymine (T) Guanine (G) Cytosine (C)
Gene Edit System
Undo
Redo
Step 1 / 1
ACGGTTAGGTACGATTGTCACAAATGCACGGGCGCCAATTAACTTTCGGTTAACTAGTAGCGCGCAGGCCCGCCGCAGTGGC
Save modification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v1.0.0
Lab build
Dashboard
Dr. Prince
Genomics Lab
Total DNA Sequences
40
Across 5 organisms
Total Patients
1,200
Active registry
Mutation Count
6,578
All recorded mutations
Risk Statistics
617 High
315 Low · 268 Medium
Featured sequence — BRCA1 sample 1
BRCA1
Highest risk patients
Dev Walker · PT-00002 High Risk (10)
Riya Sharma · PT-00006 High Risk (7)
Ethan Iyer · PT-00007 High Risk (6)
Kabir Patel · PT-00008 High Risk (8)
Lucas Garcia · PT-00010 High Risk (8)
Nora Verma · PT-00012 High Risk (7)

Conclusion :-

BioGazer successfully demonstrates the use of React JS in developing a modern scientific web application. The project provides an effective platform for DNA visualization, mutation tracking, gene searching, and risk analysis. Through reusable React components and a responsive user interface, the application improves the accessibility and management of genetic data while showcasing modern frontend development practices.

GitHub Repository Link :-

https://github.com/PRINCE7021/REACT_FINAL

Live Project Link :-

react-final-9ef5.vercel.app